

SAO DEMO CONTROLLER V3.0 to 3.2 – A COMPLETE SAO DEVELOPMENT HOST WITH MULTIPLE POWER OPTIONS!

POWER SOURCES

- (1) USB-C Into RP2040-Zero port (5V)
The default configuration powers/programs the SAO Demo Controller RP2040-Zero regardless of the power switch position. The USB 5V is also routed to the LiPo battery charge IC.
POWER SWITCH ON = downstream SAO is powered.
POWER SWITCH OFF = downstream SAO is not powered.
 - (2) External battery, 2-4 AA/AAA or 1S LiPo/LiPoly
(A) The default configuration is for non-rechargeable Primary/Alkaline cells with both the charger and LiPo protector disabled. Allowed input is 3 to 16 V per limits of Buck-Boost 3.3V Converter.
Practically, 3 to 8 "AA" or "AAA" batteries may be used including non-rechargeable Lithium-Iron Disulfide (Li-FeS₂) consumer batteries such as Energizer Ultimate Lithium which may offer up to 1.8V per cell when fresh. A variety of coin cells may be used as long as they can deliver enough current and do not exceed 16V.
(B) Optionally, to use a 1S LiPo/LiPoly cell verify the connection polarity to JST-PH BATT. port. (Adafruit) POWER SWITCH ON = RP2040-Zero and SAO are powered. POWER SWITCH OFF = RP2040-Zero and SAO are not powered.
 - (3) SAO input port 3.3V expected, 2 to 6V allowed
(A) The default configuration expects power to be supplied through the SAO port from the badge or host. As such, solder jumper SJ_REV_PWR_BDG "Reverse Power Badge" is open by default.
(B) As long as the badge/host does NOT have it's own power source, you may optionally provide 3.3V to the badge/host, AKA "reverse powering it" by removing reverse protection diode D4 and closing solder jumper SJ_REV_PWR_BDG. In this configuration, power is provided from your external battery or USB C into the RP2040-Zero through the 3.3V Buck-Boost Converter and into your badge or host device.
- Diodes prevent backfeeding between four input power options. Diodes may be removed and solder-bridged to improve efficiency if you make sure you don't create a backfeed power issue between the three sources.
- All power sources, with the exception of the USB-C 5V, flow through a switch for convenience. If USB-C power is supplied, the battery charger and RP2040-Zero will always be powered. The switch enables/disables the 3.3V Buck/Boost Converter. SO YOUR SAO WILL NOT GET POWER UNLESS THAT SWITCH IS ON AS WELL!

LIPO PROTECTOR

LiPo is protected by disconnecting -BATT from ground.

LIPO/LI-Ion CHARGER

For 1S LiPo or Li-Ion cells only.

POWER REV. POLARITY & BACKFEED PROTECTION

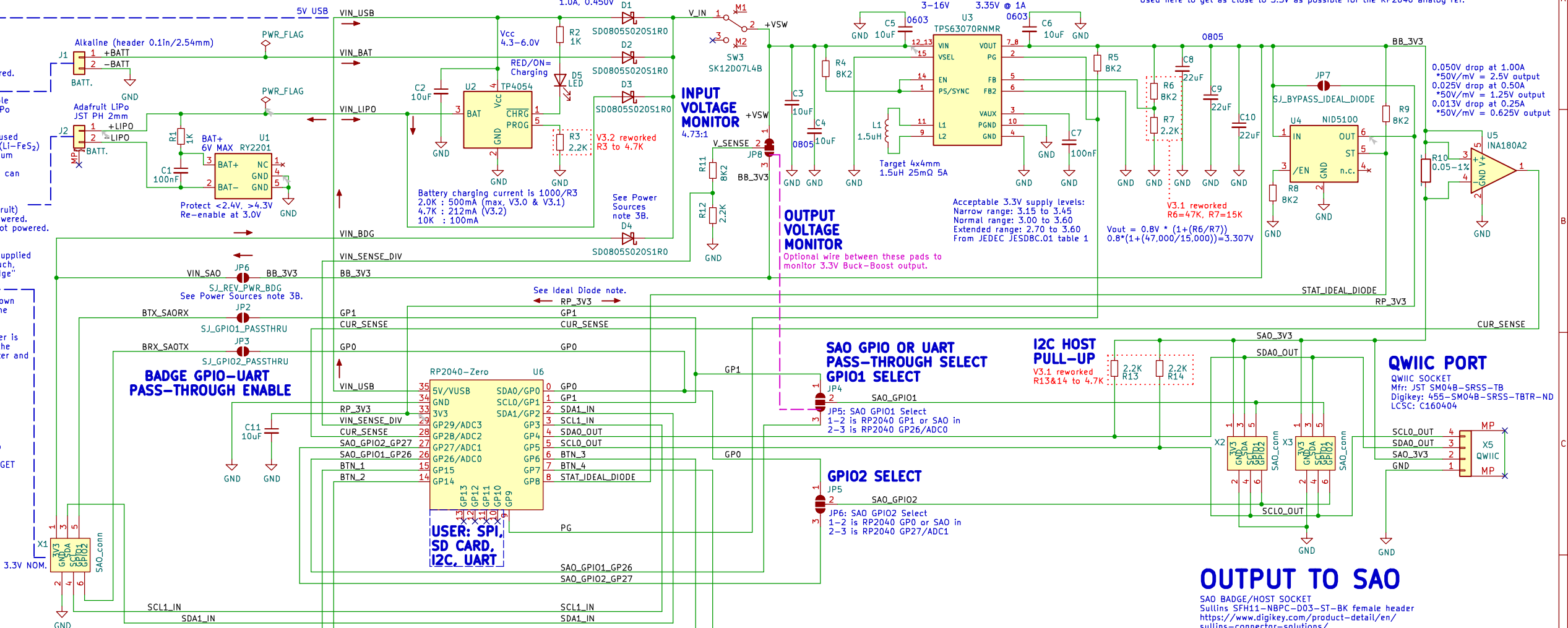
MAIN POWER SWITCH

3.3V BUCK-BOOST CONVERTER FOR RP2040-ZERO & SAO STANDARD

IDEAL DIODE

Prevents contention between RP2040-Zero 3V3 LDO output if USB C is connected. If USB C is not connected, 3V3 is supplied to the RP2040 from the external buck-boost converter. Used here to get as close to 3.3V as possible for the RP2040 analog ref.

CURRENT SENSE AMPLIFIER



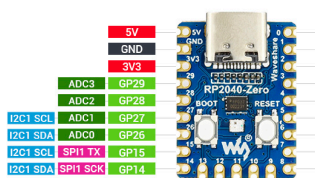
FROM HOST BADGE

SAO CLIENT PLUG
<https://www.digikey.com/en/products/detail/adam-tech/BHR-06-VUA/10414837>

GPIO1 is also data from the badge to the SAO (Badge TX, SAO RX).
GPIO2 is also data from the SAO to the badge (Badge RX, SAO TX).

MICROCONTROLLER: RP2040-ZERO

DEMO CONTROLLER FUNCTIONS:
INPUT 3-5V from Badge or USB-C
GROUND
3V3 @ ~200mA regulated from ZERO
VIN/2 input voltage measurement
SAO output current measurement
SAO GPIO2
SAO GPIO1
USER BUTTON 2
USER BUTTON 1



RP2040-Zero
See <https://www.waveshare.com/wiki/RP2040-Zero> for more information and more GPIO on bottom pads.
Onboard buttons not user accessible.
GPIO16 is NeoPixel onboard.

DEMO CONTROLLER FUNCTIONS:
SAO GPIO2 pass-through or Badge RX / SAO TX
SAO GPIO1 pass-through or Badge TX / SAO RX
SDA1 IN from Badge
SCL1 IN from Badge
SDA0 OUT to SAO
SCL0 OUT to SAO
USER BUTTON 3
USER BUTTON 4
USER 5100 STATUS/UART
GPIO/I2C/UART
USER SPI_CLK/I2C1_SDA/UART
USER SPI_TX_OUT/I2C1_SCL/UART
USER SPI_RX_IN/I2C0_SDA/UART
USER SPI_CS_N/I2C0_SCL/UART

COMMON I2C ADDRESSES

OLED (SSD1306) 128x32 0x3C
OLED (SSD1306) 128x64 0x3D
OLED (SSD1327) 128x128 0x3C

SAO STANDARDS

PRIMARY: <https://docs.google.com/document/d/1EJqyKLMAPsQ9VWF5A4eWol0qMKYr5Giw5rqRmtmM/edit?usp=sharing>
SUPPLEMENT: <https://hackaday.io/project/175182-simple-add-ons-sao>

QWIIC/STEMMA RESOURCES

<https://learn.adafruit.com/introducing-adafruit-stemma-qt/what-is-stemma>

OUTPUT TO SAO

SAO BADGE/HOST SOCKET
Sullins SFH11-NBPC-D03-ST-BK female header
<https://www.digikey.com/product-detail/en/sullins-connector-solutions/SFH11-NBPC-D03-ST-BK/S9717-ND/4558818>

REVISION HISTORY

- V3.0 (built in July 2026)
* Was not distributed. These were all reworked to at least V3.1.
- V3.1 (distributed through August 3, 2026)
* Buck-Boost output voltage setting corrected to 3.3V by changing R6/R7 from 8.2K/2.2K to 4.7K/15K.
* I2C Pull-ups R13&14 reworked from 2.2K to 4.7K for more reliably I2C operation.
* Charging circuit has 2.2K current program resistor, 545 mA. This is a little high for the included LiPo pack. Suggest replacing that resistor with 4.7K for 212mA charge current limit.
* May or may not have V3.1 label added to the board on the top of the USB-C connector.
- V3.2 (started distribution August 15, 2026)
* Reduced LiPo charging current, reworked R3 from 2.2K (454 mA) to 4.7K (212mA)
* Add V3.2 label.

All non-polarized capacitors are X7R or X5R ceramic unless otherwise noted.

Design by Andy Geppert @ www.MachineIdeas.com

Sheet: /
File: SAO_Demo_Controller.kicad_sch

Title: SAO Demo Controller, Micro Badge Development Board

Size: B	Date: 2026-08-15	Rev: 3.0, 3.1, 3.2
KiCad E.D.A. 10.0.1		Id: 1/1

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